

UniMap-Verify: One-Class Map-Image Consistency Verification with MLLM-Guided Semantic Hard Negatives

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Abstract—Ensuring consistency between maps and the real world is critical, yet manual verification remains time-consuming. Aerial imagery offers a cost-effective and widely available proxy for large-scale map verification. In this work, we address unsupervised / one-class map-image consistency verification, where training relies solely on aligned positive pairs without any annotated mismatches. While realistic for map maintenance, this setting is challenging because random negative samples are often too trivial, failing to guide the model toward fine-grained cross-modal verification. To address this problem, we propose UniMap-Verify, a task-focused framework that performs Multimodal Large Language Model (MLLM)-guided semantic hard negative mining. Specifically, we use structured semantic descriptors generated by a MLLM to construct negative samples that are semantically similar to the positives yet geometrically inconsistent with them. A curriculum schedule progressively increases the difficulty of these negatives during training. This strategy shifts the learning focus from coarse semantic shortcuts to geometry-sensitive verification. A lightweight prototype branch is further introduced as an auxiliary regularizer on the joint embedding space. Additionally, we introduce MapCheck-Bench, a unified benchmark built from three public sources, comprising approximately 49k triplets of aerial images, maps, and corresponding MLLM-generated semantic descriptions spanning 42 unique cities or regions. Extensive experiments demonstrate that UniMap-Verify significantly outperforms existing unsupervised baselines under the positive-pair-only protocol. Our code and dataset will be publicly released at <https://github.com/yangbokang/UniMap-Verify>.

Index Terms—Map-Aerial imagery consistency verification, One-class learning, Positive-pair-only learning, Semantic-guided hard negative mining

I. INTRODUCTION

WITH rapid urbanization, maps play a crucial role in providing a reference framework that links the physical world to virtual environments. To remain reliable, maps must be continuously updated as changes occur in the real world [1], [2]. However, due to the high cost and labor intensity of manual surveying and verification, map updates are typically performed only in response to substantial changes, resulting in notable delays between real-world evolution and its reflection in map data. The increasing availability of high-resolution aerial imagery offers a scalable and cost-effective means of observing the current state of the world, making it possible to support map maintenance with computer vision techniques [3]. This gives rise to a fundamental challenge: how to auto-

matically verify whether existing map data remains consistent with the latest aerial imagery.

We formulate this problem as *unsupervised / one-class map-image consistency verification*, which aims to determine whether a given map-image pair is consistent. In this work, we focus on building footprint verification, as urban evolution is primarily manifested through the construction and demolition of buildings [4]. This formulation reflects realistic data conditions in map maintenance: matched (positive) pairs corresponding to “no change” are abundant, whereas outdated or incorrect (negative) samples are sparse, unpredictable, and costly to annotate at scale. As a result, learning exclusively from positive pairs under a positive-pair-only setting becomes both natural and necessary.

Designing a robust consistency metric in this setting is highly challenging due to the substantial *modality gap* between aerial imagery and maps. Aerial images contain rich textures, illumination variations, and fine-grained visual details, whereas maps represent geographic entities in a sparse, symbolic, and highly abstract manner [5]. While human observers can bridge this gap through high-level semantic reasoning—such as recognizing the shape of a stadium despite abstract representation—enabling such reasoning within a learning framework without explicit supervision remains challenging.

More precisely, we adopt a *one-class* training protocol: the model only observes aligned map-image pairs during training, and all mismatched pairs used for optimization are synthesized online as synthetic negatives rather than collected as annotated change examples. We use the term “unsupervised” only in this restricted sense, namely that no negative labels or change masks are provided during training. This task definition is therefore stricter than conventional weakly supervised change detection: training data contain only positive pairs, while negatives appear only at evaluation time or as online synthetic mismatches.

A critical examination of existing literature highlights the necessity of a one-class strategy. First, although “heterogeneous image” matching is well-studied (e.g., optical vs. SAR) [6]–[8], the modality gap in those tasks is qualitatively different from the symbolic-to-real gap between maps and aerial imagery, where the latter must bridge abstract geometric primitives with continuous visual signals. Second, map-image matching models [9], [10] typically rely on supervised settings that require large-scale datasets of both matching and non-matching pairs. This requirement creates a bottleneck, as it ignores the real-world scenario where “changed” (negative) examples are rare. Therefore, there is an urgent need for a framework capable of learning robust consistency metrics

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solely from aligned, positive data.

Within this setting, contrastive learning has emerged as a dominant solution for change detection [11]–[14]. However, in a one-class verification setting, the core challenge is no longer simply designing a better encoder; it is constructing synthetic negatives that match the target failure cases. Conventional contrastive frameworks typically rely on random negative sampling. When applied to map–image verification, these negatives are often *overly easy*, exhibiting obvious semantic mismatches (e.g., pairing an urban map with forest imagery). Models trained in this manner tend to exploit coarse semantic or appearance *shortcuts* rather than learning genuine geometric correspondence. As a result, they generalize poorly to challenging cases where the semantics are similar, but the fine-grained geometry is inconsistent.

To address these limitations, we propose UniMap-Verify, a task-focused framework explicitly designed around semantic hard negatives. Our key insight is that effective one-class verification requires distinguishing pairs that are semantically similar but geometrically inconsistent. We therefore use the zero-shot semantic understanding capability of a Multimodal Large Language Model (MLLM) to extract structured semantic attributes and guide MLLM-driven hard negative mining. On top of that, we introduce a curriculum schedule that progressively increases the proportion of semantic hard negatives, shifting learning from coarse semantic cues to fine-grained geometric discrepancies. A prototype branch is retained only as an auxiliary regularizer, rather than the main source of supervision.

In addition, we introduce MapCheck-Bench, a unified benchmark that combines three public datasets and enriches them with MLLM-generated semantic descriptors. This benchmark enables rigorous evaluation across diverse urban patterns and geographic regions.

Our contributions are summarized as follows:

- We formalize *unsupervised / one-class map-image consistency verification*, a positive-pair-only task in which training uses only aligned map-image pairs and learns from online synthetic negatives instead of annotated mismatches.
- We propose UniMap-Verify, whose main methodological contribution is MLLM-guided semantic hard negative mining for constructing semantically similar but geometrically inconsistent training pairs.
- We introduce a curriculum schedule that progressively increases hard-negative difficulty during training, and we use a prototype branch only as an auxiliary regularizer rather than the main learning objective.
- We introduce MapCheck-Bench, a large-scale benchmark built from three public datasets and comprising over 49,000 triplets of maps, aerial images, and corresponding semantic descriptions across 42 unique cities/regions.

II. RELATED WORK

Recent research increasingly frames automatic map updating as an incremental refinement problem rather than full reconstruction from scratch. The MUNO21 dataset exemplifies

this shift, supporting operations such as adding, removing, and modifying roads while preserving validated regions [10]. Other works have focused on vector-image alignment and annotation correction, including deep learning methods for cadastral misalignment correction [15] and the detection of missing buildings in OSM [16]. General vector-map alignment approaches based on optimization and self-supervision have also been proposed [17]. With the rise of foundation models (e.g., SAM [18]), zero-shot segmentation has enabled unsupervised discrepancy detection between optical imagery and OpenStreetMap (OSM), scaling map maintenance without extensive supervision [17]. Meanwhile, comprehensive surveys and advanced methods on road extraction and vectorization provide essential methodological context [19], [20]. Collectively, these works establish a foundation for discriminative frameworks trained only on matching pairs by introducing strategies for alignment, error correction, and handling incomplete annotations.

Unsupervised change detection has also contributed significantly to this area, specifically in tackling the challenges of heterogeneous-sensor analysis, such as optical-SAR data comparison. Many approaches learn modality-invariant representations: for instance, structural relationship graphs with graph-convolutional autoencoders (SR-GCAE) capture structural similarity in multimodal pairs [21], while Markovian co-segmentation jointly infers change masks across heterogeneous modalities [7]. Other methods follow a harmonize-then-compare strategy, using cross-modal translation (with change-aware priors to ignore changed regions [22]) or specialized fusion networks [23] to achieve harmonization. Self-supervised contrastive learning is also widely applied: weighted contrastive losses with context-aware Transformers alleviate change leakage in SAR [8], radar-optical masked autoencoders learn transferable bimodal features [24], and semantic-to-change triplets explicitly model temporal differences [25]. While effective for cross-sensor change detection, these approaches are not designed for the map update problem, where the challenge lies in bridging the geometric and semantic gaps between maps and remote sensing imagery.

III. METHODOLOGY

In this section, we present UniMap-Verify, a task-focused framework for unsupervised / one-class map-image consistency verification. Given a map-image pair, the goal is to determine whether they are mutually consistent, without access to labeled negative samples during training. As illustrated in Figure 1, UniMap-Verify adopts a dual-stream architecture to encode map and image inputs separately, then centers learning on MLLM-guided semantic hard negative mining with curriculum scheduling, and finally applies a discriminative consistency objective supplemented by an auxiliary prototype regularizer.

Given aligned map-image pairs, where both inputs are rasterized as RGB patches, $x_{map} \in \mathbb{R}^{H \times W \times 3}$ represents the map layer and $x_{img} \in \mathbb{R}^{H \times W \times 3}$ is the aerial imagery, we aim to learn a scoring function $S(x_{map}, x_{img}) \in \mathbb{R}$ such that S is high for corresponding pairs and low for non-corresponding ones. During training, non-corresponding pairs

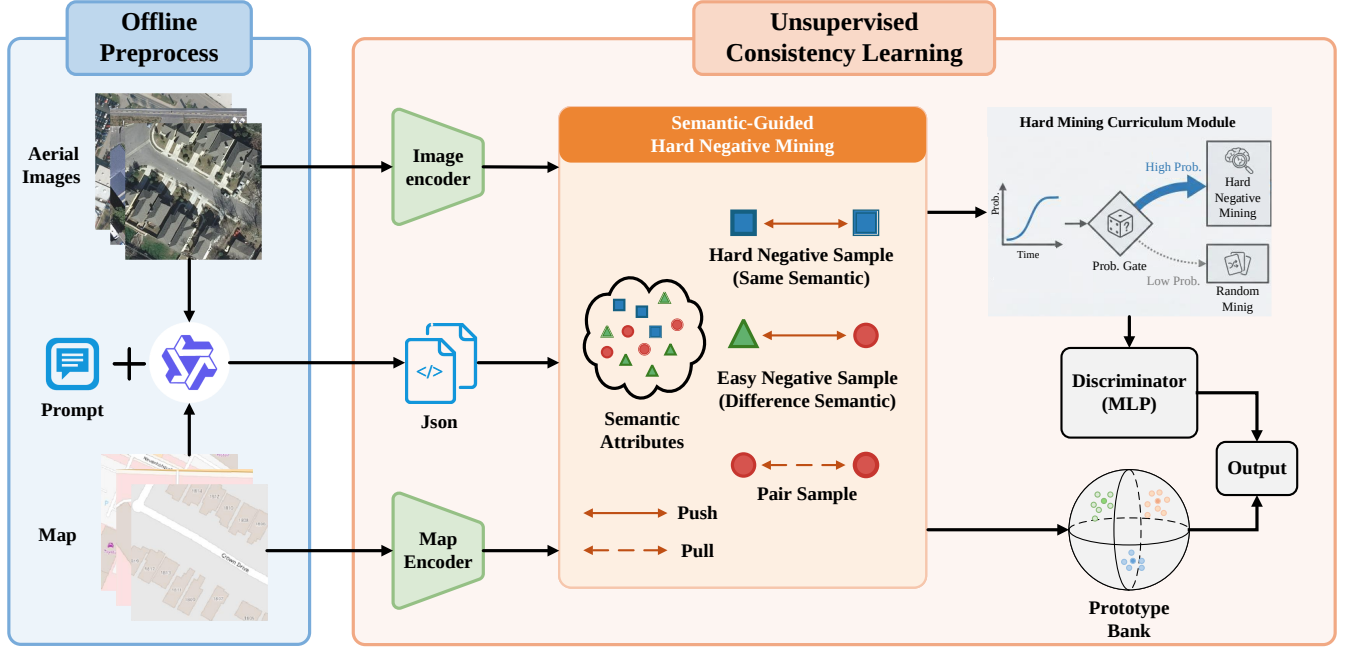


Fig. 1. Overview of UniMap-Verify. The framework is centered on one-class map-image consistency verification with MLLM-guided semantic hard negative mining. Negatives are sampled according to semantic attributes derived from MLLM-generated descriptions, and a curriculum schedule progressively increases the proportion of within-group hard negatives. A discriminator branch provides the main verification signal, while the prototype branch acts only as an auxiliary regularizer.

are not annotated data samples; they are synthetic negatives formed by re-pairing matched samples within a mini-batch. This positive-pair-only protocol makes negative construction the central source of supervision.

A. Feature Extraction and Fusion

We use a shared backbone (pre-trained on ImageNet) to extract multi-scale features from both the map image x_{map} and the aerial image x_{img} . Although maps and aerial images are heterogeneous, they share mid-level geometric cues such as footprint boundaries, block layout, and local spatial repetition. Tying the backbone weights encourages both modalities to be projected into a compatible representation space while avoiding unnecessary parameter duplication. Features are extracted from intermediate layers (specifically Layer 2 and Layer 3) to capture both structural detail and scene semantics; relying only on the final stage would discard the fine-grained geometric information needed for footprint verification.

To handle spatial variability, we employ a patch-based processing module. The extracted feature maps are unfolded into patches, interpolated to a fixed reference size, and then processed via an adaptive average pooling layer to obtain compact embeddings $f_{map} \in \mathbb{R}^D$ and $f_{img} \in \mathbb{R}^D$.

These heterogeneous features are then fused via concatenation and projected into a joint embedding space:

$$f_{joint} = \text{Projection}(\text{Concat}(f_{map}, f_{img})) \quad (1)$$

where the Projection module consists of a multi-layer perceptron (MLP) that maps the 2D-dimensional concatenated input to the target embedding dimension D_{embed} .

B. Semantic-Guided Hard Negative Mining

In a one-class setting, synthetic negative construction becomes the main source of supervisory signal. Standard contrastive learning methods typically generate negative pairs via random batch shuffling. This often results in *trivial negatives* where the map and aerial image differ significantly in semantic context (e.g., a high-density urban map paired with a forest image), allowing the model to learn simple statistical shortcuts rather than fine-grained geometric alignment.

To address this, we introduce MLLM-guided Semantic Hard Negative Mining, which is the primary mechanism of UniMap-Verify. We utilize a pre-trained MLLM to generate a structured JSON descriptor for each training sample with four keys: `density_class`, `urban_style`, `geographic_context`, and a free-text description. The first three fields are directly converted into a semantic label tuple $l_i = (d_i, u_i, g_i)$, where d_i , u_i , and g_i denote `density_class`, `urban_style`, and `geographic_context`, respectively. The free-text description is retained only for qualitative analysis and dataset inspection. Invalid or incomplete outputs are mapped to an unknown group for robustness. This deterministic tuple-based construction avoids an additional embedding-and-clustering stage and makes the grouping rule fully reproducible.

During training, given a batch of B positive pairs $\{(x_{map}^i, x_{img}^i)\}_{i=1}^B$, we construct synthetic negative pairs (x_{map}^i, x_{img}^j) ($i \neq j$). We employ a curriculum learning strategy to sample index j . Specifically, with probability $p_{hard}(t)$, we sample j from the same semantic group as i (i.e., $l_i = l_j$), creating a *hard negative* that shares high-level semantics but

Algorithm 1 Training Procedure of UniMap-Verify

Require: Aligned map-image pairs with semantic labels $\mathcal{D} = \{(x_{map}^i, x_{img}^i, l_i)\}$, total epochs T

Ensure: Trained encoders, projection head, discriminator, and prototypes

Initialize E_{map} , E_{img} , Projection head, Discriminator D , and Prototypes P

for epoch $t = 1$ to T **do**

Update curriculum probability $p_{hard}(t)$ via Eq. (2)

for each batch $B \in \mathcal{D}$ **do**

Step 1: Feature Encoding

$f_{map}^i \leftarrow E_{map}(x_{map}^i)$, $f_{img}^i \leftarrow E_{img}(x_{img}^i)$

$f_{joint}^{real,i} \leftarrow \text{Projection}(\text{Concat}(f_{map}^i, f_{img}^i))$

Step 2: Semantic-Guided Negative Sampling

Identify candidate set $\mathcal{H}_i = \{k \in B \mid l_k = l_i, k \neq i\}$

Sample negative index j :

if $\mathcal{H}_i \neq \emptyset$ **and** $\text{Rand}() < p_{hard}(t)$ **then**

Sample $j \sim \text{Uniform}(\mathcal{H}_i)$ // *Hard Negative*

else

Sample $j \sim \text{Uniform}(\{k \in B \mid k \neq i\})$ //

Random Negative

end if

$f_{img}^j \leftarrow E_{img}(x_{img}^j)$

$f_{joint}^{fake,i} \leftarrow \text{Projection}(\text{Concat}(f_{map}^i, f_{img}^j))$

Step 3: Dual-Branch Optimization

Compute $\mathcal{L}_{disc}$ and $\mathcal{L}_{proto}$

$\mathcal{L}_{total} \leftarrow \mathcal{L}_{disc} + \lambda_{proto} \cdot \mathcal{L}_{proto}$

Update model parameters $\theta \leftarrow \theta - \eta \nabla \mathcal{L}_{total}$

end for

end for

lacks geometric correspondence. With probability $1 - p_{hard}(t)$, we sample j randomly from the batch to maintain global separability. If no same-group candidate exists in the current mini-batch, the sampler falls back to a random negative, as summarized in Algorithm 1.

The curriculum probability p_{hard} increases linearly during training:

$$p_{hard}(t) = 0.2 + 0.7 \cdot \frac{t-1}{T-1}, \quad t = 1, \dots, T \quad (2)$$

where t is the current epoch and T is the total epochs. This schedule increases $p_{hard}(t)$ linearly from 0.2 at the first epoch to 0.9 at the final epoch. This curriculum is a key training mechanism: it first stabilizes learning with easier mismatches and then progressively forces the model to focus on harder, geometry-sensitive distinctions.

C. Consistency Learning with Auxiliary Prototype Regularization

To learn a robust metric for verifying image-map consistency, we use a primary Discriminator Branch together with an auxiliary Prototype Branch.

Discriminator Branch The discriminator $D(\cdot)$ is an MLP designed to distinguish between real (positive) pairs and fake

(negative) pairs. We employ a Hinge Loss to optimize the discriminator:

$$\mathcal{L}_{disc} = \frac{1}{B} \sum_{i=1}^B \left(\max(0, m_{disc} - D(f_{joint}^{real,i})) + \max(0, m_{disc} + D(f_{joint}^{fake,i})) \right) \quad (3)$$

where $f_{joint}^{real,i}$ is the fused feature of the matched pair (x_{map}^i, x_{img}^i) , and $f_{joint}^{fake,i}$ is the fused feature of the negative pair (x_{map}^i, x_{img}^j) . m_{disc} is a margin hyperparameter. This loss encourages the discriminator to output high scores for consistent pairs and low scores for inconsistent ones.

Prototype Branch To regularize the embedding geometry beyond pairwise discrimination, we introduce a learnable Prototype Bank $P \in \mathbb{R}^{K \times D_{embed}}$, containing K prototype vectors. This branch is auxiliary: it encourages features from real pairs to stay close to a compact set of latent consistency patterns, while features from fake pairs are pushed away from such patterns.

We calculate the cosine similarity between the fused feature f_{joint} and the prototypes:

$$S_{proto}(f) = \max_k \left(\frac{f \cdot p_k}{\|f\| \|p_k\|} \right) \quad (4)$$

The prototype loss is defined as:

$$\mathcal{L}_{proto} = \mathcal{L}_{pos} + \mathcal{L}_{neg} \quad (5)$$

where:

$$\mathcal{L}_{pos} = \frac{1}{B} \sum_{i=1}^B (1 - S_{proto}(f_{joint}^{real,i})) \quad (6)$$

$$\mathcal{L}_{neg} = \frac{1}{B} \sum_{i=1}^B \text{ReLU}(S_{proto}(f_{joint}^{fake,i}) - m_{proto}) \quad (7)$$

Here, $\mathcal{L}_{pos}$ pulls positive pairs towards their nearest prototype, while $\mathcal{L}_{neg}$ pushes negative pairs away if their similarity to any prototype exceeds a margin m_{proto} .

This branch is intentionally different from a standard triplet loss. Triplet formulations impose sample-to-sample constraints anchored on a particular negative instance, whereas our prototype branch imposes sample-to-structure constraints through a shared prototype bank. It therefore complements, rather than defines, the verifier: the discriminator focuses on matched-versus-mismatched separation, while the prototype branch organizes the joint embedding space around a small set of reusable consistency patterns. Because this branch is auxiliary and jointly optimized with the discriminator, trivial concentration onto a single prototype is disfavored in practice: if hard negatives and positives were mapped to the same prototype, $\mathcal{L}_{neg}$ would increase and the fused score in Eq. (9) would deteriorate. We thus use the prototype bank as a lightweight structural regularizer rather than as a balanced clustering objective.

The total loss is a weighted sum of the losses from both branches:

$$\mathcal{L}_{total} = \mathcal{L}_{disc} + \lambda_{proto} \mathcal{L}_{proto} \quad (8)$$

(a) Geographic coverage



(b) Sample Data



(c) Semantic diversity

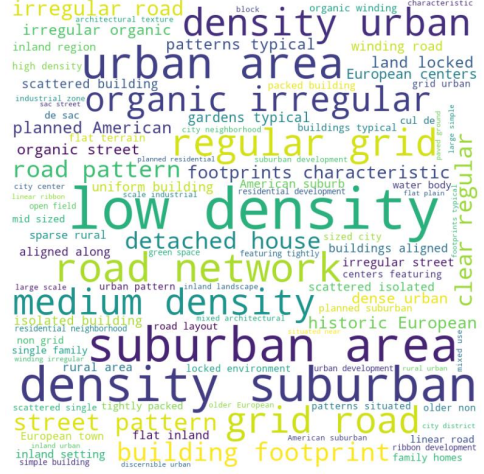


Fig. 2. Overview of MapCheck-Bench. The benchmark features (a) geographic coverage across 42 unique cities/regions, (b) sample triplets enriched with MLLM-generated semantic descriptions, and (c) a word cloud illustrating the diverse semantic attributes.

D. Inference

During inference, we compute a consistency score for a given image-map pair. The final score is a weighted fusion of the primary discriminator probability and the auxiliary similarity from the prototype bank:

$$\text{Score} = \alpha \cdot \sigma(D(f_{\text{joint}})) + (1 - \alpha) \cdot \frac{S_{\text{proto}}(f_{\text{joint}}) + 1}{2} \quad (9)$$

where σ is the sigmoid function, and α is a balancing hyperparameter. A higher score indicates a higher probability that the map and aerial image are consistent. This fusion strategy preserves the discriminator as the main verification signal while using the prototype score only as a secondary stabilizing cue.

TABLE I
DETAILED STATISTICS OF MAPCHECK-BENCH. TRAIN/VAL/TEST COUNTS FOLLOW THE SPLIT PROTOCOL DESCRIBED IN SECTION IV-C.

Subset	Res.	Train	Val	Test
Inria	0.3m	19,504	2,438	2,438
WHU	0.2m	1,920	240	240
OpenMapCD	0.25-0.5m	17,850	2,231	2,231

IV. THE MAPCHECK-BENCH BENCHMARK

To robustly evaluate the one-class map verification task, we constructed a unified benchmark, MapCheck-Bench. Rather than collecting raw imagery from scratch, our benchmark contribution lies in unifying heterogeneous public sources into a common verification task with consistent OSM pairing, semantic enrichment, split protocol, and manual test-set cleaning. This section details the data collection, annotation, and statistical distribution.

A. Data Collection

Our dataset curation aims for high geographic diversity to ensure the model’s robustness across different urban topologies. After deduplicating overlaps between subsets, the collected data spans 42 unique cities/regions across six continents including North America, Europe, Asia, Africa, South America, and Oceania. To intuitively visualize the geographic coverage, we plot the distribution of these cities on a world map, as shown in Figure 2(a).

The aerial imagery originates from three primary sources, distinguished by resolution and sensor type: (1) **Inria Subset** [26] covers 5 regions (e.g., Austin, Vienna) with high-resolution (0.3m) optical imagery; (2) **WHU Subset** [27] covers Christchurch, New Zealand, featuring 0.2m aerial imagery capturing post-earthquake reconstruction; (3) **OpenMapCD Subset** [9] covers 40 diverse regions globally, utilizing OpenEarthMap [28] and 0.25-0.5m ground sampling distance.

Crucially, regardless of the image source, all corresponding vector map data are consistently sourced from OSM [29], ensuring the benchmark reflects the real-world challenges of standardizing open-source map data.

B. Data Annotation

The annotation process consists of two stages: geometric ground truth generation and semantic description generation.

Change Map Generation. To enable quantitative evaluation, we establish pixel-level “Change Maps” that indicate discrepancies between the map and the image (i.e., changed buildings). For Inria and WHU subsets, we first collect two types of building masks that correspond to aerial imagery and map, respectively. After the collection, two types of building masks are compared in order to label changed buildings between images and maps. For OpenMapCD, we utilize existing discrepancy masks between maps and images. Patch-level changed/unchanged labels are then derived from these masks: a patch is labeled as *changed* if its change mask contains at least one changed-building pixel, and *unchanged* otherwise. All raw data were cropped into non-overlapping patches of 256×256 pixels. Finally, to ensure the reliability of the evaluation, we performed manual verification on the test set to correct obvious alignment errors or labeling noise.

Semantic Text Generation. Beyond geometric labels, we enrich the dataset with fine-grained semantic descriptions. We employ Qwen3-VL-8B to generate a structured JSON descriptor for each image patch. The prompt extracts four fields: *Density Class*, *Urban Style*, *Geographic Context*, and a short free-text *Description*. In the released benchmark, the categorical fields cover six density categories (*sparse_rural*, *low_density_suburban*, *medium_density*, *high_density_urban*, *very_high_vertical*, *large_scale_industrial*), six urban-style categories (*regular_grid*, *organic_irregular*, *linear_ribbon*, *cul_de_sac*, *scattered_isolated*, *large_scale_industrial*), and four geographic-context categories (*inland_urban*, *coastal_riparian*, *flat_plains*, *mountainous_hilly*). For hard negative mining, we use the tuple of the three categorical fields as the semantic group ID, while the free-text

description is reserved for qualitative analysis and word-cloud visualization. Figure 2(b) showcases sample data pairs alongside their generated semantic descriptions, and Appendix V-E qualitatively compares descriptors produced by different candidate MLLMs in Figure 6.

C. Data Statistics

Based on resolution and data provenance, MapCheck-Bench is organized into three subsets: Inria, WHU, and OpenMapCD. Split (Train/Val/Test) ratios are set to approximately 8 : 1 : 1 across all subsets. Table I summarizes the subset-level statistics. To rigorously adhere to the one-class, positive-pair-only verification supervision:

- **Training Set:** The raw data contain both changed and unchanged pairs. However, for our task we exclusively utilize the unchanged (positive) pairs, ensuring the model strictly learns from aligned data and never observes annotated “change” examples during training.
- **Validation & Test Sets:** These sets contain a balanced ratio ($\approx 1 : 1$) of changed (negative) and unchanged (positive) pairs. They are used only for model selection and final evaluation, not for constructing the training supervision.

Additionally, to visualize the semantic diversity of our generated text, we present a Word Cloud in Figure 2(c). Terms such as “dense,” “residential,” “grid,” and “vegetation” appear frequently, reflecting the dataset’s coverage of complex urban environments.

V. EXPERIMENTS

A. Experimental Settings

1) Implementation Details: All experiments were implemented using PyTorch on a single NVIDIA RTX 5880 Ada GPU. We used WideResNet-50 [30] pre-trained on ImageNet as the backbone for both image and map encoders, as it provides a stable compromise between receptive field and local-detail preservation for 256×256 verification patches. The models were trained for 50 epochs using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-2} . The batch size was set to 128. The hard-negative sampling probability $p_{hard}(t)$ increased linearly from 0.2 to 0.9 according to Eq. (2), and the fusion weight in Eq. (9) was set to $\alpha = 0.7$ unless otherwise noted. We use Qwen3-VL-8B [31] to generate semantic descriptions. All methods were trained and evaluated under the same one-class, positive-pair-only protocol, patch partition, and score aggregation procedure for fair comparison.

From an efficiency perspective, semantic-guided hard negative mining introduces no extra backbone forward pass. After a single encoding of each mini-batch, hard negatives are formed by re-indexing aerial-image embeddings within the batch while reusing the already computed map embeddings. Consequently, the dominant cost remains the shared backbone encoding and patch aggregation, and the additional cost of hard negative mining is negligible compared with standard batch-shuffling contrastive training. The only extra preprocessing

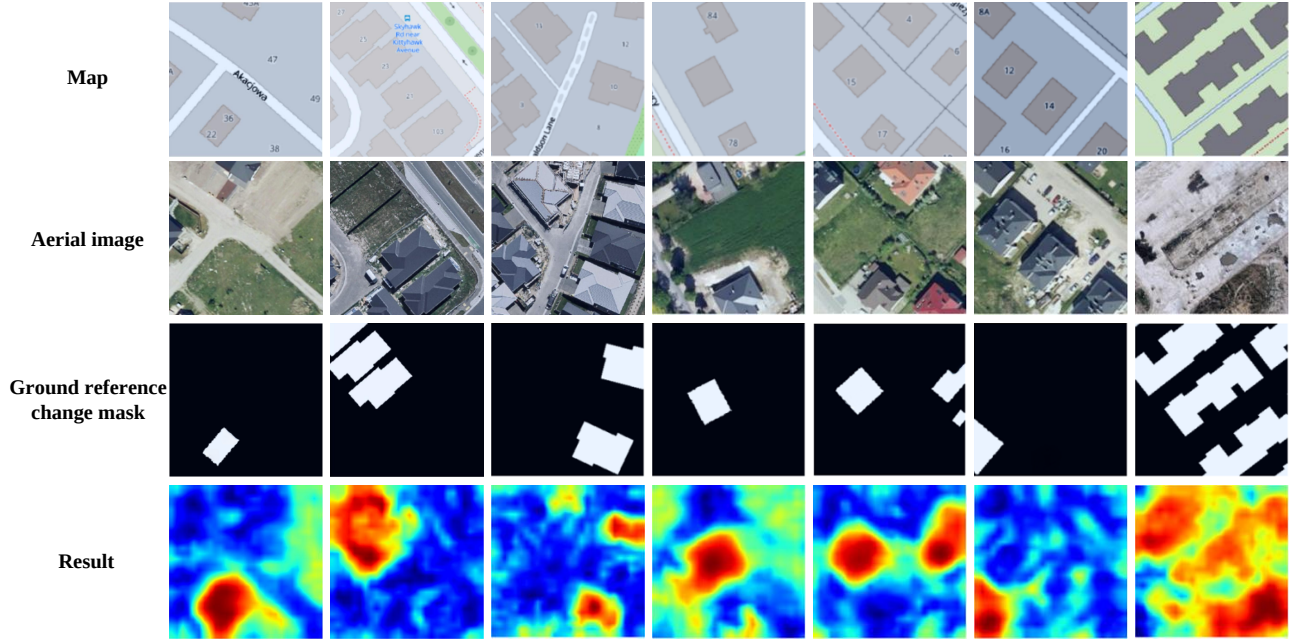


Fig. 3. Visualization of the detection results. The first row shows the map, the second row shows the aerial image, the third row shows the ground-reference change mask, and the fourth row shows the predicted result.

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS. WE REPORT AUROC (%) AND AUPR (%) ON THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Method	Inria		WHU		OpenMapCD	
	AUROC	AUPR	AUROC	AUPR	AUROC	AUPR
<i>Unsupervised Heterogeneous CD</i>						
CACD	52.42	53.11	57.39	57.16	47.40	44.68
CDRL	52.69	54.85	46.04	47.80	67.27	63.03
CFRL	48.23	50.11	41.79	43.61	49.66	46.77
PRBCD	50.77	52.91	43.57	45.70	65.59	61.27
<i>Supervised Image-Map CD (Adapted)</i>						
ObjFormer	50.26	50.84	55.08	51.54	53.48	51.19
ChangeFormer	50.16	53.20	57.98	56.87	61.04	56.86
SEIFNet	51.19	52.33	57.25	58.94	38.24	41.20
SNUNet-CD	48.05	49.44	54.07	53.35	50.00	49.43
<i>Proposed Methods</i>						
UniMap-Verify	61.71	65.32	77.72	80.42	74.73	77.15

expense is semantic descriptor generation, which is performed offline once per dataset version and reused throughout training and deployment; the MLLM is not invoked at inference time.

2) *Evaluation Metrics*: We employ two standard metrics to evaluate the performance of verification tasks: Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPR). Higher values indicate better performance in distinguishing matching pairs from non-matching ones.

B. Baselines

We compare UniMap-Verify with a comprehensive set of state-of-the-art methods categorized as follows:

- **Unsupervised Heterogeneous Change Detection**: Methods including CDRL [32], CACD [33], CFRL [34],

and PRBCD [35], which are designed to operate across different remote sensing modalities without ground truth supervision.

- **Supervised Image-Map Change Detection**: Methods such as ObjFormer [9], ChangeFormer [36], SEIFNet [37], and SNUNet-CD [38]. To adapt them to our one-class verification setting, we train each baseline using only unchanged map-image pairs. For methods that originally output pixel-level change maps, we retain their feature extraction and decoding architectures and supervise them with all-zero change masks on unchanged pairs; at evaluation time, the predicted change-probability map is spatially aggregated into an image-level anomaly score. For SNUNet-CD, we append global average pooling and a binary classification head to the final-layer feature maps, train the head to predict the normal class on

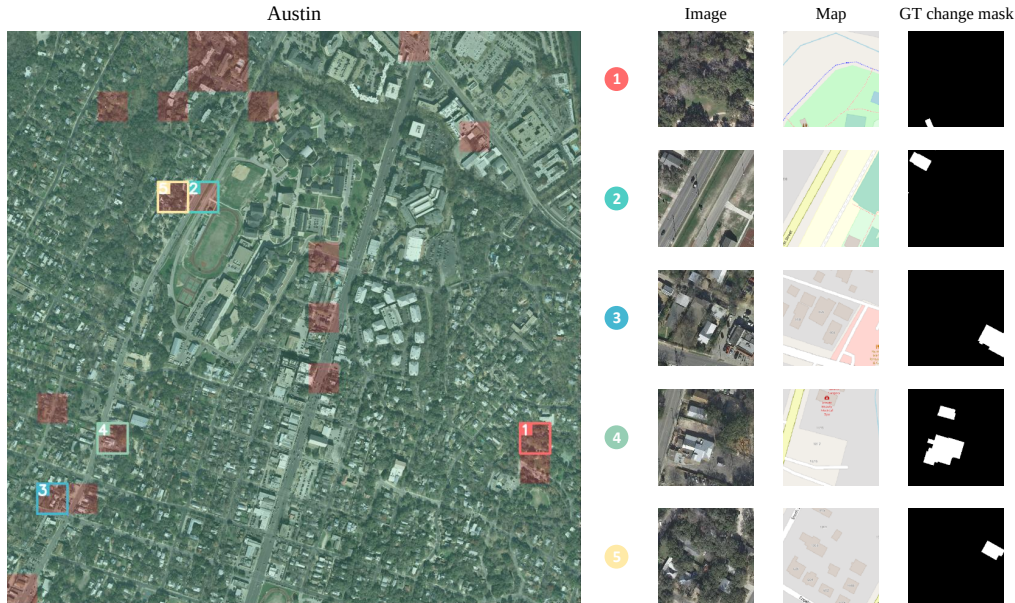


Fig. 4. Example of consistency score prediction results in an area of the city of Austin (2,250,000 m^2). Green color indicates high consistency scores, and red color indicates low consistency scores.

unchanged pairs, and use the complement of the normal probability as the anomaly score. Changed pairs are used only for validation and testing, and all methods are evaluated using AUROC and AUPR.

C. Results

As Figure 3 illustrates, UniMap-Verify effectively identifies inconsistent regions between maps and aerial images. Moreover, Figures 4 and 5 showcase the model’s capability to process large-scale urban areas in different geographic settings. Table II summarizes the quantitative comparison results under the one-class, positive-pair-only protocol.

Comparison with Unsupervised Methods. Our method significantly outperforms existing unsupervised heterogeneous change detection methods (CACD, CDRL, CFRL). For instance, on MapCheck-OpenMapCD, the full UniMap-Verify model achieves an AUROC of 74.73%, surpassing the best unsupervised baseline (CDRL) by over 7 percentage points. This supports the task-focused design: when training is restricted to positive pairs only, MLLM-guided semantic hard negatives provide a substantially stronger signal than generic unsupervised matching objectives.

Comparison with Adapted Supervised Architectures. UniMap-Verify also outperforms several supervised architectures (e.g., ChangeFormer, ObjFormer) after adapting them to the same positive-pair-only protocol. This suggests that MLLM-guided hard negative mining is more important to this task than simply transplanting conventional change-detection architectures into a one-class verification setting.

D. Ablation Study

To verify the effectiveness of the proposed components, we conducted a comprehensive ablation study covering three key

TABLE III
ABLATION STUDY. EFFECTIVENESS OF MLLM-GUIDED SEMANTIC HARD NEGATIVE MINING, AUXILIARY PROTOTYPE REGULARIZATION, AND THE CURRICULUM-BASED HARD NEGATIVE SCHEDULE.

Configuration	Inria	WHU	OpenMapCD
	AUROC, AUPR	AUROC, AUPR	AUROC, AUPR
Full Model	61.71, 65.32	77.72, 80.42	74.73, 77.15
w/o MLLM	59.70, 60.98	75.66, 79.54	69.53, 68.99
w/o PT	62.15, 65.92	77.61, 80.12	75.11 , 76.84
w/o Curr.	60.12, 63.10	76.80, 79.90	73.10, 75.10

aspects: (1) MLLM-guided semantic hard negative mining, (2) auxiliary prototype regularization, and (3) the curriculum-based hard negative schedule.

As the results in Table III indicate, removing the MLLM semantic descriptors (“w/o MLLM”) leads to a clear performance drop on all datasets, particularly on MapCheck-OpenMapCD (-5.20% AUROC and -8.16% AUPR). This confirms that MLLM-guided hard negative mining is the primary source of gain over random batch sampling in the one-class setting.

The Prototype Branch (“w/o PT”) has a mixed effect and therefore should not be presented as a uniformly beneficial module. On Inria, removing PT slightly improves both AUROC and AUPR (+0.44% / +0.60%). On WHU, the full model remains better on both metrics. On OpenMapCD, “w/o PT” achieves a slightly higher AUROC (75.11% vs. 74.73%), whereas the full model achieves a higher AUPR (77.15% vs. 76.84%). Therefore, PT does not provide a consistent gain across WHU and OpenMapCD in the strict metric-by-metric sense; instead, it behaves as an auxiliary regularizer whose value depends on dataset composition. Inria covers



Fig. 5. Example of consistency score prediction results in a local area in Christchurch. Green color indicates high consistency scores, and red color indicates low consistency scores.

only 5 regions and has narrower geographic diversity, so the discriminator trained with semantic hard negatives is already strong and PT can mildly over-regularize. OpenMapCD spans 40 regions globally and poses the strongest cross-domain generalization requirement, where PT improves ranking quality in AUPR even though AUROC is slightly lower. WHU contains only one city, but its post-earthquake reconstruction scenario induces a domain shift different from Inria, and PT still yields a small positive effect there.

Finally, removing the curriculum-based schedule (“w/o Curr.”) reduces performance on the verified Inria and OpenMapCD results reported in Table III, indicating that the curriculum is an important training mechanism. The WHU entry for this ablation is left as *TBD* pending result verification, so we avoid drawing a dataset-specific conclusion from an unverified number. Taken together, the available evidence still supports the role of curriculum learning in gradually increasing hard-negative difficulty so that the model can first learn coarse alignment before focusing on fine-grained verification.

The inference score in UniMap-Verify is a weighted fusion of the Discriminator probability and the Prototype similarity, controlled by α (Eq. (9)). To validate this auxiliary fusion design, we evaluated the model performance on the WHU dataset with varying $\alpha \in [0, 1]$.

As shown in Table IV, the best performance is achieved at $\alpha = 0.7$, indicating that the discriminative head contributes the majority of the signal while the prototype head provides a complementary auxiliary cue. Using either branch alone ($\alpha = 0$ or $\alpha = 1$) results in suboptimal performance on WHU, confirming that the prototype score is useful but secondary.

TABLE IV
SENSITIVITY ANALYSIS OF α ON THE WHU DATASET.

α	0.0	0.1	0.3	0.5	0.6	0.7	0.8	0.9	1.0
Branch	Prototype-Dominant					Fused	Discriminator-Dominant		
AUROC	71.80	73.15	75.40	76.85	77.21	77.72	77.58	76.90	77.61

E. Qualitative Comparison of Descriptor Generators

Our Semantic-Guided Hard Negative Mining relies on MLLM-generated semantic descriptors to cluster samples and construct semantically similar but geometrically mismatched negatives. We therefore compare four candidate MLLMs qualitatively: InternVL3-8B, Qwen2.5-VL-3B, Qwen2.5-VL-7B, and Qwen3-VL-8B. As illustrated in Figure 6, Qwen3-VL-8B tends to produce more specific and spatially grounded descriptors, whereas the smaller models more often collapse to generic urban labels. This qualitative comparison motivates the text-level analysis below; however, we do not claim a conclusive downstream verification ranking here because this paper does not include a dedicated descriptor-generator performance table.

F. Text-Level Semantic Quality

Beyond qualitative inspection, we directly analyzed the quality of the generated JSONL descriptors on the Inria dataset (24,380 samples). Figure 7 and Table V summarize three metrics: (1) **Schema Compliance** (rate of valid JSON parsing), (2) **Descriptive Detail** (average word count per description), and (3) **Lexical Diversity** (unique vocabulary size).

As summarized in Table V, we observed the following:

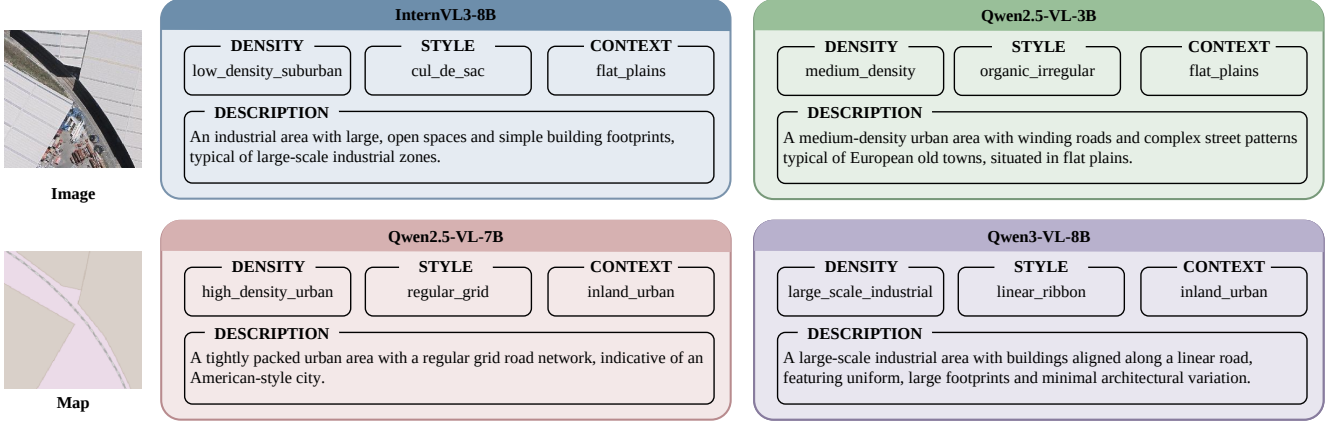


Fig. 6. Comparison of MLLM-generated semantic descriptions. We compare InternVL3-8B, Qwen2.5-VL-3B, Qwen2.5-VL-7B, and Qwen3-VL-8B on a sample pair.

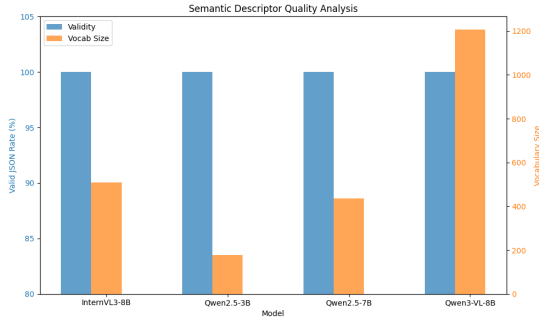


Fig. 7. Statistical analysis of generated semantic descriptors.

TABLE V
STATISTICAL ANALYSIS OF GENERATED SEMANTIC DESCRIPTORS ON INRIA DATASET. WHILE ALL MODELS ADHERE STRICTLY TO THE OUTPUT SCHEMA, QWEN3-VL-8B EXHIBITS SIGNIFICANTLY HIGHER LEXICAL DIVERSITY AND DESCRIPTION LENGTH.

Model	Valid JSON (%)	Avg. Length (words)	Vocab Size
InternVL3-8B	100.00%	22.45	509
Qwen2.5-VL-3B	100.00%	19.97	178
Qwen2.5-VL-7B	100.00%	18.84	436
Qwen3-VL-8B	100.00%	24.16	1,206

- **High Schema Adherence:** All four models achieved a 100% valid JSON rate, indicating that our prompt template is robust and compatible across different MLLM architectures.
- **Superior Lexical Diversity:** The most striking difference lies in vocabulary size. Qwen3-VL-8B utilizes a vocabulary of 1,206 unique words, which is nearly $2.4\times$ that of InternVL3-8B (509) and $6.7\times$ that of Qwen2.5-VL-3B (178). This suggests that smaller or less capable models tend to reuse a limited set of generic terms (e.g., repeating “dense residential” for all city blocks), whereas Qwen3-VL-8B uses a much richer vocabulary to describe subtle variations in urban texture, vegetation, and style.
- **Richness of Detail:** Qwen3-VL-8B also generates the

longest descriptions (24.16 words/sample), providing more detailed context than Qwen2.5-VL-7B (18.84 words), which further supports the effectiveness of our semantic-guided hard negative mining.

G. Rationale for Selecting Qwen3-VL-8B

Based on the qualitative comparison in Figure 6 and the text-level statistics in Figure 7 and Table V, we selected **Qwen3-VL-8B** as the default generator for the following reasons:

- 1) **Richer Semantic Specificity:** It yields the largest vocabulary and the longest descriptions, indicating better coverage of subtle urban attributes that are useful for semantic grouping.
- 2) **No Robustness Trade-off:** It maintains the same 100% valid-JSON rate as the other candidates, so the added descriptive richness does not come at the cost of schema reliability.

VI. CONCLUSION

Keeping maps consistent with the evolving physical world is crucial for modern navigation, yet remains costly to maintain. We introduced UniMap-Verify for unsupervised / one-class map-image consistency verification under a strict positive-pair-only training protocol. The central contribution is MLLM-guided semantic hard negative mining, together with a curriculum schedule that progressively increases training difficulty; the prototype branch plays only an auxiliary regularization role. We also presented MapCheck-Bench, a large-scale benchmark with rich semantic annotations to facilitate rigorous evaluation, including 49k map-image-text pairs across 42 unique cities/regions. Extensive experiments on the proposed benchmark demonstrate that UniMap-Verify significantly outperforms existing unsupervised baselines in this one-class setting. While this work focuses on building-footprint verification, the semantic hard-negative framework is generic and can be extended to other map elements such as roads, water systems, or land use. Future work may explore integrating the verification output with automatic map correction systems, moving from discrepancy detection to direct map repair.

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